FUTURES

Learning Outcome

- define forward contracts, futures contracts, swaps, options (calls and puts), and credit derivatives and compare their basic characteristics

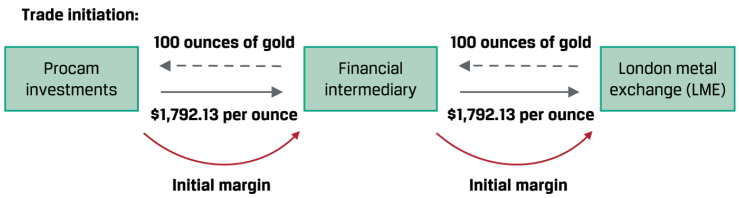
Futures contracts are forward contracts with standardized sizes, dates, and underlyings that trade on futures exchanges. Futures markets offer both greater liquidity and protection against loss by default by combining contract uniformity with an organized market with rules, regulations, and a central clearing facility.

The futures contract buyer creates a long exposure to the underlying by agreeing to purchase the underlying at a later date at a pre-agreed price. The seller makes the opposite commitment, creating a short exposure to the underlying by agreeing to sell the underlying asset in the future at an agreed-on price. This agreed-on price is called the **futures price**, $f_0(T)$. The frequency of futures contract maturities, contract sizes, and other details are established by the exchange based on buyer and seller interest.

The most important feature of futures contracts is the daily settlement of gains and losses and the associated credit guarantee provided by the exchange through its clearinghouse. At the end of each day, the clearinghouse engages in a practice called **mark to market (MTM)**, also known as the **daily settlement**. The clearinghouse determines an average of the final futures trading price of the day and designates that price as the end-of-day **settlement price**. All contracts are then said to be marked to the end-of-day settlement price.

As with forward contracts, no cash is exchanged when a futures contract is initiated by a buyer or seller. However, each counterparty must deposit a required minimum sum (or **initial margin**) into a **futures margin account** held at the exchange that is used by the clearinghouse to settle the daily mark to market. Futures contracts must be executed with specialized financial intermediaries that clear and settle payments at the exchange on behalf of counterparties, as shown in Example 2.

EXAMPLE 2



As in Example 1, Procam Investments enters a cash-settled contract to buy 100 ounces of gold at a price of $1,792.13 per ounce in three months. Instead of the forward in Example 1, Procam purchases a futures contract [$f_0(T)$ = $1,792.13] on the exchange via a financial intermediary. London Metal Exchange rules require an initial cash margin of $4,950 per gold contract (100 ounces) sold or purchased:

- Procam deposits $4,950 in required initial margin with the exchange.
- Today's spot gold price (S_0) is $1,770 per ounce, and the opening gold futures price, $f_0(T)$, is $1,792.13 per ounce.
- At today's close, the gold futures price, $f_1(T)$, settles at $1,797.13 per ounce.

- Procam's futures margin account has an ending balance for the day of $5,450, or $4,950 initial margin plus the $500 MTM gain.

Each futures contract specifies a **maintenance margin**, or minimum balance set below the initial margin, that each contract buyer and seller must hold in the futures margin account from trade initiation until final settlement at maturity. The clearinghouse moves funds daily between the buyer and seller margin accounts, crediting the accounts of those with mark to market gains and charging those with mark-to-market losses.

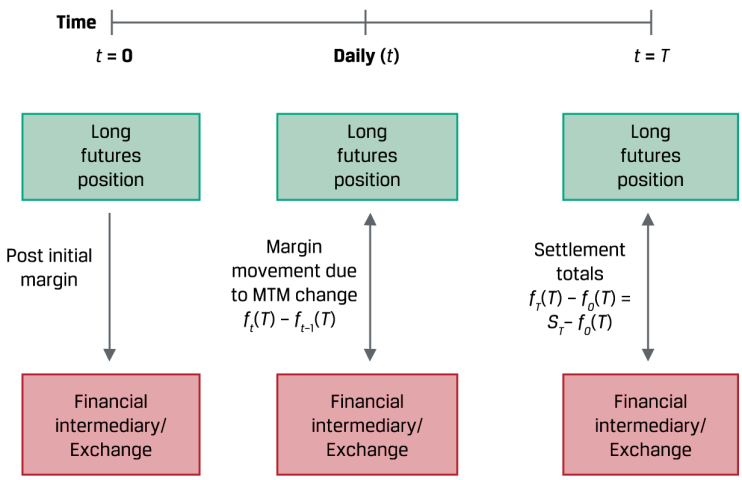
For example, London Metal Exchange rules require a maintenance margin of $4,500 per 100-ounce gold contract sold or purchased. Now consider the *seller* of a futures contract with a position that offsets that of Procam in Example 2.

- Seller deposits $4,950 in required initial margin with the exchange.
- Today's opening gold futures price, $f_0(T)$, is $1,792.13 per ounce.
- At the close, the gold futures price, $f_1(T)$, settles at $1,797.13 per ounce.
- Seller realizes a $500 MTM loss, or $5 per ounce × 100 ounces; $500 is deducted from its futures margin account by the clearinghouse.
- Seller's futures margin account has an ending balance of $4,450, or $4,950 initial margin less the $500 MTM loss.
- Seller's margin account is $50 below the required maintenance margin ($4,450 – $4,500) = –$50

The seller receives a **margin call**, or request to immediately deposit funds to return the account balance to the *initial margin*. The seller must deposit $500 in order to bring the margin account back to the $4,950 initial margin. The amount required to replenish the futures margin account is sometimes referred to as **variation margin**. If a counterparty fails to meet the margin call, it must close out the contract as soon as possible and cover any additional losses. If the counterparty cannot meet its obligations, the clearinghouse provides a guarantee that it will cover the loss itself by maintaining an insurance fund. Exhibit 2 shows the futures margining and settlement process, where $f_0(T)$ is the futures price at inception and $f_t(T)$ represents the futures price on day t.

Exhibit 2:

Futures Margin and Settlement Process



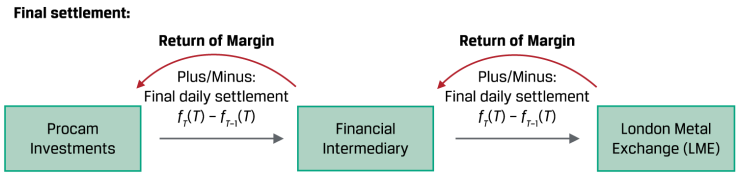
▶ Expand / collapse Extended Description

Exchanges reserve the right to impose more strict requirements than standard futures margin account rules to limit potential losses from counterparty default. For example, for large positions or a significant increase in price volatility of the underlying, an exchange

a band relative to the previous day's settlement price within which all trades must occur. If market participants wish to trade at a price outside these bands, trading stops until two parties agree on a trade at a price within the prescribed range. In other cases, exchanges use what is called a **circuit breaker** to pause intraday trading for a brief period if a price limit is reached.

Similar to forward contracts, final settlement at maturity for futures contracts is based on the difference between the futures price, $f_0(T)$, and the underlying price of S_T, or $[S_T - f_0(T)]$ from the buyer's perspective. Because Procam has agreed to purchase gold now (at $t = T$) valued at $1,780.50 per ounce at a price of $1,792.13 per ounce, it owes $11.63 per ounce, or $1,163 [($1,780.50 – $1,792.13) × 100 oz.] under both the forward from Example 1 and this futures contract. The *net* payoff profile shown in Exhibit 1 is the same for a futures contract as for a forward assuming they have the same maturity date, with the difference being the *timing* of the cash flows due to the daily futures contract mark-to-market settlement, as shown in Example 3.

EXAMPLE 3



For purposes of exposition, we compress the three months of gold futures price changes from Example 2 into six days of trading in the following spreadsheet:

Exhibit 3

Procam's Futures Margin Account		
Gold Contract	100	ounces
# of Contracts	1	
Initial Futures Price $f_0(T)$	$1,792.13	per ounce
Initial Position Value	$179,213	
Initial Margin	$4,950	
Maintenance Margin	$4,500	

Day	Futures Price	Day Gain (Loss)	Total Gain (Loss)	Margin Balance	Margin Call
$T-6$	$1,792.13			$4,950	
$T-5$	$1,797.13	$500	$500	$5,450	—
$T-4$	$1,786.25	($1,088)	($588)	$4,362	$588
$T-3$	$1,782.19	($406)	($994)	$4,544	—
$T-2$	$1,777.45	($474)	($1,468)	$4,070	$880
$T-1$	$1,779.50	$205	($1,263)	$5,155	—
T	$1,780.50	$100	**($1,163)**	$5,255	$1,468

Procam's Results from the Futures Contract	
Gold Contract	100
# of Contracts	1
Initial Futures Price $f_0(T)$	$1,792.13

Total Gain (Loss)	($1,163)
Final Futures Price $F_T(T)$	$1,780.50
Final Position Value	$178,050
Sum of Margin Calls	$1,468
Beginning less Ending Margin Balance	($305)
Total Payments to Margin Account	$1,163

As the gold futures price, $f_0(t)$, falls and the margin balance drops below the $4,500 maintenance margin minimum over time, Procam must immediately replenish its balance to the $4,950 initial margin each time this occurs:

- On Day T – 4, gold futures fall $10.88 per ounce ($1,797.13 – $1,786.25). Procam's balance falls $1,088 (100 × $10.88) to $4,362 ($5,450 – $1,088). Procam faces a margin call of $588 ($4,950 – $4,362).
- On Day T – 2, gold futures fall $4.74 per ounce ($1,782.19 – $1,777.45). Procam's balance falls $474 (100 × $4.74) to $4,070 ($4,544 – $474). Procam faces a margin call of $880 ($4,950 – $4,070).

On the final trading day, Procam has paid a total of $1,468 ($588 + $880) in margin calls and its futures margin account balance is $5,155, or $205 in excess of the $4,950 initial margin. Procam has a cumulative MTM loss of $1,263 ($1,468 – $205) at the start of the last trading day.

- The prior day's gold futures settlement price, $f_{T-1}(T)$, is $1,779.50.
- Gold futures rise $1 per ounce on the final trading day to settle at $f_T(T)$ = $1,780.50 per ounce, the same as for the forward in Example 1.
- The daily change in Procam's margin account is an increase of $100 ($1 per ounce × 100 ounces), bringing the margin account to $5,255.
- Procam's futures margin balance of $5,255 is returned at settlement for a net return of $305 ($5,255 – $4,950) in margin.
- Procam receives a net return of $305 in margin at settlement for a cumulative loss upon settlement of $1,163 ($305 – $1,468).

Both the forward and futures contracts involve a $1,163 settlement loss, but the forward is fully settled at maturity while the futures contract is settled based on the daily MTM. The time value of money principle suggests that these forward and futures settlements are not equivalent amounts of money, but the differences are small for shorter maturities and low interest rates. Also note that under the forward contract in Example 1, the financial intermediary bears counterparty risk to Procam for the forward settlement. In practice, financial intermediaries often use collateral arrangements similar to futures margining for forwards or other derivatives to reduce counterparty credit risk.

At maturity, the number of outstanding contracts, or **open interest**, is settled via cash or physical delivery. A counterparty may instead choose to enter an offsetting futures contract before expiration to close out a position; for example, a futures contract buyer may simply sell the open contract. The clearinghouse marks the contract to the current price relative to the previous settlement price and closes out the participant's position.

Futures contracts specify whether physical delivery of an underlying or cash settlement occurs at expiration. For example, a commodity futures contract with physical delivery obligates the seller to deliver an underlying asset of a specific type, amount, and quality to a designated location. The buyer must accept and pay for delivery, which ensures that the futures price converges with the spot price at expiration.

Question Set: Futures

margin account falls below the _____ _____, or minimum balance that each contract buyer and seller must hold in the account from trade initiation until final settlement, the buyer must immediately deposit funds to return the account balance to the _____ _____.

Hide Solution

Solution:

If a futures contract buyer's margin account falls below the *maintenance margin*, or minimum balance that each contract buyer and seller must hold in the account from trade initiation until final settlement, the buyer must immediately deposit funds to return the account balance to the *initial margin*.

2. Describe the mark-to-market process for a futures contract.

Hide Solution

Solution:

The exchange clearinghouse determines an average of the final futures prices of the day and designates that price as the end-of-day settlement price. The daily settlement of gains and losses takes place via each counterparty's futures margin account.

3. Identify these futures contract participants that correspond to the following statements:

A Futures contract purchaser	1. Must make a margin deposit at contract initiation and maintain a minimum balance until maturity
B. Futures contract seller	2. Receives a margin account deposit if the futures price increases on any trading day
C. Both a futures contract purchaser and a seller	3. Receives a positive payoff if the spot price S_T is below the futures price, $f_0(T)$, at maturity

Hide Solution

Solution:

1. The correct answer is C. Both futures contract buyers and sellers must deposit margin and maintain a minimum margin balance (maintenance margin) over the life of a contract.

2. The correct answer is A. The futures contract buyer realizes a mark-to-market gain on any trading day when the futures price increases and receives a corresponding margin account deposit.

3. The correct answer is B. The futures contract seller realizes a positive payoff at maturity if $S_T < f_0(T)$.

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